

Undergraduate Texts in Mathematics

Editors

S. Axler

F.W. Gehring

K.A. Ribet

Springer

New York

Berlin

Heidelberg

Barcelona

Hong Kong

London

Milan

Paris

Singapore

Tokyo

Theodore W. Gamelin

Complex Analysis

With 184 Illustrations



Springer

Theodore W. Gamelin
Department of Mathematics
UCLA
Box 951555
Los Angeles, CA 90095-1555
USA

Editorial Board

S. Axler
Mathematics Department
San Francisco State
University
San Francisco, CA 94132
USA

F.W. Gehring
Mathematics Department
East Hall
University of Michigan
Ann Arbor, MI 48109
USA

K.A. Ribet
Mathematics Department
University of California
at Berkeley
Berkeley, CA 94720-3840
USA

Mathematics Subject Classification (2000): 28-01, 30-01

Library of Congress Cataloging-in-Publication Data
Gamelin, Theodore W.

Complex analysis / Theodore W. Gamelin.

p. cm. — (Undergraduate texts in mathematics)

Includes bibliographical references and index.

ISBN 0-387-95093-1 (alk. paper) — ISBN 0-387-95069-9 (softcover : alk. paper)

1. Mathematical analysis. 2. Functions of complex variables. I. Title.

II. Series.

QA300 .G25 2000

515—dc21

00-041905

Printed on acid-free paper.

© 2001 Springer-Verlag New York, Inc.

All rights reserved. This work may not be translated or copied in whole or in part without the written permission of the publisher (Springer-Verlag New York, Inc., 175 Fifth Avenue, New York, NY 10010, USA), except for brief excerpts in connection with reviews or scholarly analysis. Use in connection with any form of information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed is forbidden.

The use of general descriptive names, trade names, trademarks, etc., in this publication, even if the former are not especially identified, is not to be taken as a sign that such names, as understood by the Trade Marks and Merchandise Marks Act, may accordingly be used freely by anyone.

Production managed by Timothy Taylor; manufacturing supervised by Erica Bresler.
Typeset by MacroTeX, Savoy, IL.

Printed and bound by R.R. Donnelley and Sons, Harrisonburg, VA.

Printed in the United States of America.

9 8 7 6 5 4 3 2 1

ISBN 0-387-95093-1

SPIN 10772447 (hardcover)

ISBN 0-387-95069-9

SPIN 10770762 (softcover)

Springer-Verlag New York Berlin Heidelberg

A member of BertelsmannSpringer Science+Business Media GmbH

To the many wonderful students at Perugia who took my course, and to the others there who contributed to an enjoyable experience

Preface

This book provides an introduction to complex analysis for students with some familiarity with complex numbers from high school. Students should be familiar with the Cartesian representation of complex numbers and with the algebra of complex numbers, that is, they should know that $i^2 = -1$. A familiarity with multivariable calculus is also required, but here the fundamental ideas are reviewed. In fact, complex analysis provides a good training ground for multivariable calculus. It allows students to consolidate their understanding of parametrized curves, tangent vectors, arc length, gradients, line integrals, independence of path, and Green's theorem. The ideas surrounding independence of path are particularly difficult for students in calculus, and they are not absorbed by most students until they are seen again in other courses.

The book consists of sixteen chapters, which are divided into three parts. The first part, Chapters I–VII, includes basic material covered in all undergraduate courses. With the exception of a few sections, this material is much the same as that covered in Cauchy's lectures, except that the emphasis on viewing functions as mappings reflects Riemann's influence. The second part, Chapters VIII–XI, bridges the nineteenth and the twentieth centuries. About half this material would be covered in a typical undergraduate course, depending upon the taste and pace of the instructor. The material on the Poisson integral is of interest to electrical engineers, while the material on hyperbolic geometry is of interest to pure mathematicians and also to high school mathematics teachers. The third part, Chapters XII–XVI, consists of a careful selection of special topics that illustrate the scope and power of complex analysis methods. These topics include Julia sets and the Mandelbrot set, Dirichlet series and the prime number theorem, and the uniformization theorem for Riemann surfaces. The final five chapters serve also to complete the coverage of all background necessary for passing PhD qualifying exams in complex analysis.

Note to the instructor

There is a glut of complex analysis textbooks on the market. It is a beautiful subject, so beautiful that a large number of experts have been moved to

write their own accounts of the area. In spite of the plethora of textbooks, I have never found an introduction to complex analysis that is completely suitable for my own teaching style and audiences.

The students in each of my various audiences have begun the course with a wide range of backgrounds. Teaching to students with disparate backgrounds and preparations has posed a major teaching challenge. I respond by including early some topics that can be treated in an elementary way and yet are usually new and capture the imagination of students with already some background in complex analysis. For example, the stereographic projection appears early, the Riemann surface of the square root function is explained early at an intuitive level, and both conformality and fractional linear transformations are treated relatively early. Exercises range from the very simple to the quite challenging, in all chapters. Some of the exercises that appear early in the book can form the basis for an introduction to a more advanced topic, which can be tossed out to the more sophisticated students. Thus for instance the basis is laid for introducing students to the spherical metric already in the first chapter, though the topic is not taken up seriously until much later, in connection with Marty's theorem in Chapter XII.

The second problem addressed by the book has to do with flexibility of use. There are many routes through complex analysis, and many instructors hold strong opinions concerning the optimal route. I address this problem by laying out the material so as to allow for substantial flexibility in the ordering of topics. The instructor can defer many topics (for instance, the stereographic projection, or conformality, or fractional linear transformations) in order to reach Cauchy's theorem and power series relatively early, and then return to the omitted topics later, time permitting.

There is also flexibility with respect to adjusting the course to undergraduate students or to beginning graduate students. The bulk of the book was written with undergraduate students in mind, and I have used various preliminary course notes for Chapters I–XI at the undergraduate level. By adjusting the level of the lectures and the pace I have found the course notes for all sixteen chapters appropriate for a first-year graduate course sequence.

One of my colleagues wrote in commenting upon the syllabus of our undergraduate complex analysis course that “fractional powers should be postponed to the end of the course as they are very difficult for the students.” My philosophy is just the reverse. If a concept is important but difficult, I prefer to introduce it early and then return to it several times, in order to give students time to absorb the idea. For example, the idea of a branch of a multivalued analytic function is very difficult for students, yet it is a central issue in complex analysis. I start early with a light introduction to the square root function. The logarithm function follows soon, followed by phase factors in connection with fractional powers. The basic idea is returned to several times throughout the course, as in the applications of

residue theory to evaluate integrals. I find that by this time most students are reasonably comfortable with the idea.

A solid core for the one-semester undergraduate course is as follows:

Chapter I

Chapter II

Sections III.1-5

Sections IV.1-6

Sections V.1-7

Sections VI.1-4

Sections VII.1-4

Sections VIII.1-2

Sections IX.1-2

Sections X.1-2

Sections XI.1-2

To reach power series faster I would recommend postponing I.3, II.6-7, III.4-5, and going light on Riemann surfaces. Sections II.6-7 and III.4-5 should be picked up again before starting Chapter IX.

Which additional sections to cover depends on the pace of the instructor and the level of the students. My own preference is to add more contour integration (Sections VII.5 and VII.8) and hyperbolic geometry (Section IX.3) to the syllabus, and then to do something more with conformal mapping, as the Schwarz reflection principle (Section X.3), time permitting. To gain time, I mention some topics (as trigonometric and hyperbolic functions) only briefly in class. Students learn this material as well by reading and doing assigned exercises. Finishing with Sections XI.1-2 closes the circle and provides a good review at the end of the term, while at the same time it points to a fundamental and nontrivial theorem (the Riemann mapping theorem).

Note to the student

You are about to enter a fascinating and wonderful world. Complex analysis is a beautiful subject, filled with broad avenues and narrow backstreets leading to intellectual excitement. Before you traverse this terrain, let me provide you with some tips and some warnings, designed to make your journey more pleasant and profitable.

Above all, give some thought to strategies for study and learning. This is easier if you are aware of the difference between the “what,” the “how,” and the “why,” (as Halmos calls them). The “what” consists of definitions, statements of theorems, and formulae. Determine which are most important and memorize them, at least in slogan form if not precisely. Just as one maintains in memory the landmark years 1066, 1453, and 1776 as markers in the continuum of history, so should you maintain in memory the definition of analytic function, the Cauchy-Riemann equations, and the residue formula. The simplest of the exercises are essentially restatements of “what.”

The “how” consists in being able to apply the formulae and techniques to solve problems, as to show that a function is analytic by checking the Cauchy-Riemann equations, or to determine whether a polynomial has a zero in a certain region by applying the argument principle, or to evaluate a definite integral by contour integration. Before determining “how” you must know “what.” Many of the exercises are “how” problems. Working these exercises and discussing them with other students and the instructor are an important part of the learning process.

The “why” consists in understanding why a theorem is true or why a technique works. This understanding can be arrived at in many different ways and at various levels. There are several things you can do to understand why a result is true. Try it out on some special cases. Make a short synopsis of the proof. See where each hypothesis is used in the proof. Try proving it after altering or removing one of the hypotheses. Analyze the proof to determine which ingredients are absolutely essential and to determine its depth and level of difficulty. The slogan form of the Jordan curve theorem is that “every closed curve has an inside and an outside” (Section VIII.7). What is the level of difficulty of this theorem? Can you come up with a direct proof? Try it.

Finally, be aware that there is a language of formal mathematics that is related to but different from common English. We all know what “near” means in common English. In the language of formal mathematics the word carries with it a specific measure of distance or proximity, which is traditionally quantified by $\varepsilon > 0$ or a “for every neighborhood” statement. Look also for words like “eventually,” “smooth,” and “local.” Prepare to absorb not only new facts and ideas but also a different language. Developing some understanding of the language is not easy – it is part of growing up and becoming mathematically sophisticated.

Acknowledgments

This book stems primarily from courses I gave in complex analysis at the Interuniversity Summer School at Perugia (Italy). Each course was based on a series of exercises, for which I developed a computer bank. Gradually I deposited written versions of my lectures in the computer bank. When I finally decided to expand the material to book form, I also used notes based on lectures presented over the years at several places, including UCLA, Brown University, Valencia (Spain), and long ago at the university at La Plata (Argentina). I have enjoyed teaching this material. I learned a lot, both about the subject matter and about teaching, through my students. I would like to thank the many students who contributed, knowingly or unknowingly, to this book.

The origins of many of the mathematical ideas have been lost in the thickets of the history of mathematics. Let me mention the source for one item. The treatment of the parabolic case of the uniformization theorem follows a line of proof due to D. Marshall, and I am grateful for his sharing

his work. As far as I know, everything else is covered by the bibliography, and I apologize for any omissions.

Each time I reread a segment of the book manuscript I found various mathematical blunders, grammatical infringements, and stylistic travesties. Undoubtedly mistakes have persisted into the printed book. I would appreciate receiving your email about any egregious errors you come across, together with your comments about any passages you perceive to be particularly dense or unenlightening. My email address, while I am around, is twg@math.ucla.edu. I thank you, dear reader, in advance.

Julie Honig and Mary Edwards helped with the preparation of class notes that were used for parts of the book, and for this I thank them. Finally, I am happy to acknowledge the skilled assistance of the publishing staff, who turned my doodles into figures and otherwise facilitated publication of the book.

T.W. Gamelin

Pacific Palisades February 2001

Contents

Preface	vii
Introduction	xvii

FIRST PART

Chapter I	The Complex Plane and Elementary Functions	1
1.	Complex Numbers	1
2.	Polar Representation	5
3.	Stereographic Projection	11
4.	The Square and Square Root Functions	15
5.	The Exponential Function	19
6.	The Logarithm Function	21
7.	Power Functions and Phase Factors	24
8.	Trigonometric and Hyperbolic Functions	29
Chapter II	Analytic Functions	33
1.	Review of Basic Analysis	33
2.	Analytic Functions	42
3.	The Cauchy-Riemann Equations	46
4.	Inverse Mappings and the Jacobian	51
5.	Harmonic Functions	54
6.	Conformal Mappings	58
7.	Fractional Linear Transformations	63
Chapter III	Line Integrals and Harmonic Functions	70
1.	Line Integrals and Green's Theorem	70
2.	Independence of Path	76
3.	Harmonic Conjugates	83
4.	The Mean Value Property	85
5.	The Maximum Principle	87
6.	Applications to Fluid Dynamics	90
7.	Other Applications to Physics	97

Chapter IV	Complex Integration and Analyticity	102
1.	Complex Line Integrals	102
2.	Fundamental Theorem of Calculus for Analytic Functions	107
3.	Cauchy's Theorem	110
4.	The Cauchy Integral Formula	113
5.	Liouville's Theorem	117
6.	Morera's Theorem	119
7.	Goursat's Theorem	123
8.	Complex Notation and Pompeiu's Formula	124
Chapter V	Power Series	130
1.	Infinite Series	130
2.	Sequences and Series of Functions	133
3.	Power Series	138
4.	Power Series Expansion of an Analytic Function	144
5.	Power Series Expansion at Infinity	149
6.	Manipulation of Power Series	151
7.	The Zeros of an Analytic Function	154
8.	Analytic Continuation	158
Chapter VI	Laurent Series and Isolated Singularities	165
1.	The Laurent Decomposition	165
2.	Isolated Singularities of an Analytic Function	171
3.	Isolated Singularity at Infinity	178
4.	Partial Fractions Decomposition	179
5.	Periodic Functions	182
6.	Fourier Series	186
Chapter VII	The Residue Calculus	195
1.	The Residue Theorem	195
2.	Integrals Featuring Rational Functions	199
3.	Integrals of Trigonometric Functions	203
4.	Integrands with Branch Points	206
5.	Fractional Residues	209
6.	Principal Values	212
7.	Jordan's Lemma	216
8.	Exterior Domains	219
SECOND PART		
Chapter VIII	The Logarithmic Integral	224
1.	The Argument Principle	224
2.	Rouché's Theorem	229
3.	Hurwitz's Theorem	231
4.	Open Mapping and Inverse Function Theorems	232
5.	Critical Points	236
6.	Winding Numbers	242

7. The Jump Theorem for Cauchy Integrals	246
8. Simply Connected Domains	252
Chapter IX The Schwarz Lemma and Hyperbolic Geometry	260
1. The Schwarz Lemma	260
2. Conformal Self-Maps of the Unit Disk	263
3. Hyperbolic Geometry	266
Chapter X Harmonic Functions and the Reflection Principle	274
1. The Poisson Integral Formula	274
2. Characterization of Harmonic Functions	280
3. The Schwarz Reflection Principle	282
Chapter XI Conformal Mapping	289
1. Mappings to the Unit Disk and Upper Half-Plane	289
2. The Riemann Mapping Theorem	294
3. The Schwarz-Christoffel Formula	296
4. Return to Fluid Dynamics	304
5. Compactness of Families of Functions	306
6. Proof of the Riemann Mapping Theorem	311
THIRD PART	
Chapter XII Compact Families of Meromorphic Functions	315
1. Marty's Theorem	315
2. Theorems of Montel and Picard	320
3. Julia Sets	324
4. Connectedness of Julia Sets	333
5. The Mandelbrot Set	338
Chapter XIII Approximation Theorems	342
1. Runge's Theorem	342
2. The Mittag-Leffler Theorem	348
3. Infinite Products	352
4. The Weierstrass Product Theorem	358
Chapter XIV Some Special Functions	361
1. The Gamma Function	361
2. Laplace Transforms	365
3. The Zeta Function	370
4. Dirichlet Series	376
5. The Prime Number Theorem	382
Chapter XV The Dirichlet Problem	390
1. Green's Formulae	390
2. Subharmonic Functions	394
3. Compactness of Families of Harmonic Functions	398
4. The Perron Method	402
5. The Riemann Mapping Theorem Revisited	406

6. Green's Function for Domains with Analytic Boundary	407
7. Green's Function for General Domains	413
Chapter XVI Riemann Surfaces	418
1. Abstract Riemann Surfaces	418
2. Harmonic Functions on a Riemann Surface	426
3. Green's Function of a Surface	429
4. Symmetry of Green's Function	434
5. Bipolar Green's Function	436
6. The Uniformization Theorem	438
7. Covering Surfaces	441
Hints and Solutions for Selected Exercises	447
References	469
List of Symbols	471
Index	473

Introduction

Complex analysis is a splendid realm within the world of mathematics, unmatched for its beauty and power. It has varifold elegant and often-times unexpected applications to virtually every part of mathematics. It is broadly applicable beyond mathematics, and in particular it provides powerful tools for the sciences and engineering.

Already in the eighteenth century Euler discovered the connection between trigonometric functions and exponential functions through complex analysis. (It was he who invented the notation $e^{i\theta}$.) However, it was not until the nineteenth century that the foundations of complex analysis were laid. Among the many mathematicians and scientists who contributed, there are three who stand out as having influenced decisively the course of development of complex analysis. The first is A. Cauchy (1789-1857), who developed the theory systematically along the lines we shall follow, with the complex integral calculus, Cauchy's theorem, and the Cauchy integral formula playing fundamental roles. The other two are K. Weierstrass (1815-1897) and B. Riemann (1826-1866), who appeared on the mathematical scene about the middle of the nineteenth century. Weierstrass developed the theory from a starting point of convergent power series, and this approach led towards more formal algebraic developments. Riemann contributed a more geometric point of view. His ideas had a tremendous impact not only on complex analysis but upon mathematics as a whole, though his views took hold only gradually.

In addition to the standard undergraduate material, we shall follow several strands and obtain several poster theorems, which together with the more elementary material cover what might be called the "complex analysis canon," the part of complex analysis included in the syllabus of the typical PhD qualifying exam.

One of the strands we shall follow culminates in the prime number theorem. Already Euler in the eighteenth century had written down an infinite product for the zeta function, connecting the prime numbers to complex analysis. In the 1830's Dirichlet used variants of the zeta function to prove the existence of infinitely many primes in arithmetic progressions. Riemann

did fundamental work connecting the zeta function to the distribution of prime numbers. And finally just before the close of the nineteenth century J. Hadamard and C.J. de la Vallée Poussin independently proved the prime number theorem using techniques of complex analysis.

Another strand we shall follow is the conformal mapping of domains in the plane and more generally of Riemann surfaces. We shall aim at two poster results: the Riemann mapping theorem and the uniformization theorem for Riemann surfaces. The definitive version of the Riemann mapping theorem, which one finds in all complex analysis textbooks today, was proved by W. Osgood in 1900. The uniformization theorem for Riemann surfaces was proved independently in 1907 by P. Koebe and H. Poincaré, thereby solving Hilbert's 22nd problem from his famous address to the International Mathematical Congress in 1900.

The first quarter of the twentieth century was one of rapid development of the foundations of complex analysis. P. Montel put his finger on the notion of compactness in spaces of meromorphic functions and developed the theory of normal families. P. Fatou and G. Julia used Montel's theorem in their seminal work around 1914-1921 on complex iteration theory. On another front, O. Perron developed in 1923 a powerful method for solving the Dirichlet problem.

By the end of the first quarter of the twentieth century, the complex analysis canon had been established, and nearly all the main results constituting the undergraduate and first-year graduate courses in complex analysis had been obtained. Nevertheless, throughout the twentieth century there has been much exciting progress on the frontiers of research in complex analysis, and meanwhile proofs of the most difficult foundational results have been gradually simplified and clarified. While the complex analysis canon has remained relatively static, the developments at the frontier have led to new perspectives and shifting emphases. For instance, the current research interest in dynamical systems and the advent of computer graphics contributed to elevating the work of Fatou and Julia to a more prominent position.

What lies before you is the distillation of the essential, the useful, and the beautiful, from two centuries of labor. Enjoy!

I

The Complex Plane and Elementary Functions

In this chapter we set the scene and introduce some of the main characters. We begin with the three representations of complex numbers: the Cartesian representation, the polar representation, and the spherical representation. Then we introduce the basic functions encountered in complex analysis: the exponential function, the logarithm function, power functions, and trigonometric functions. We view several concrete functions $w = f(z)$ as mappings from the z -plane to the w -plane, and we consider the problem of describing the inverse functions.

1. Complex Numbers

A **complex number** is an expression of the form $z = x + iy$, where x and y are real numbers. The component x is called the **real part** of z , and y is the **imaginary part** of z . We will denote these by

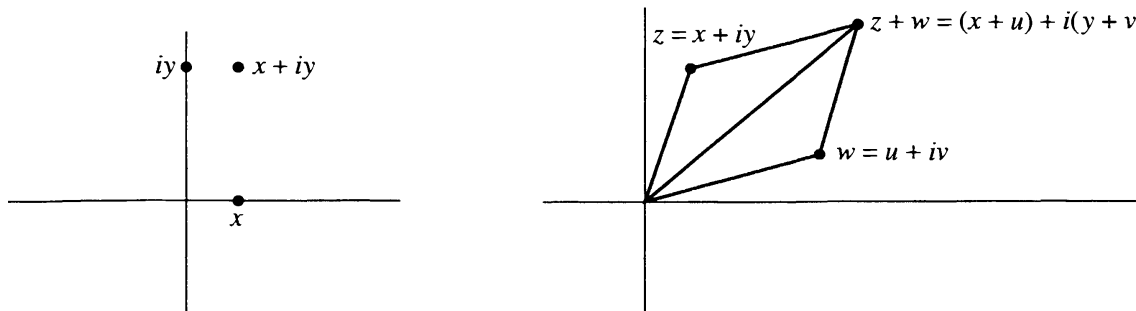
$$\begin{aligned}x &= \operatorname{Re} z, \\y &= \operatorname{Im} z.\end{aligned}$$

The set of complex numbers forms the **complex plane**, which we denote by $\mathbb{C}$. We denote the set of real numbers by $\mathbb{R}$, and we think of the real numbers as being a subset of the complex plane, consisting of the complex numbers with imaginary part equal to zero.

The correspondence

$$z = x + iy \longleftrightarrow (x, y)$$

is a one-to-one correspondence between complex numbers and points (or vectors) in the Euclidean plane $\mathbb{R}^2$. The real numbers correspond to the x -axis in the Euclidean plane. The complex numbers of the form iy are called **purely imaginary numbers**. They form the **imaginary axis** $i\mathbb{R}$



in the complex plane, which corresponds to the y -axis in the Euclidean plane.

We add complex numbers by adding their real and imaginary parts:

$$(x + iy) + (u + iv) = (x + u) + i(y + v).$$

Thus $\operatorname{Re}(z + w) = \operatorname{Re}(z) + \operatorname{Re}(w)$, and $\operatorname{Im}(z + w) = \operatorname{Im}(z) + \operatorname{Im}(w)$ for $z, w \in \mathbb{C}$. The addition of complex numbers corresponds to the usual componentwise addition in the Euclidean plane.

The **modulus** of a complex number $z = x + iy$ is the length $\sqrt{x^2 + y^2}$ of the corresponding vector (x, y) in the Euclidean plane. The modulus of z is also called the **absolute value** of z , and it is denoted by $|z|$:

$$|z| = \sqrt{x^2 + y^2}.$$

The triangle inequality for vectors in the plane takes the form

$$|z + w| \leq |z| + |w|, \quad z, w \in \mathbb{C}.$$

By applying the triangle inequality to $z = (z - w) + w$, we obtain $|z| \leq |z - w| + |w|$. Subtracting $|w|$, we obtain a very useful inequality,

$$(1.1) \quad |z - w| \geq |z| - |w|, \quad z, w \in \mathbb{C}.$$

Complex numbers can be multiplied, and this is the feature that distinguishes the complex plane $\mathbb{C}$ from the Euclidean plane $\mathbb{R}^2$. Formally, the multiplication is defined by

$$(x + iy)(u + iv) = xu - yv + i(xv + yu).$$

One can check directly from this definition that the usual laws of algebra hold for complex multiplication:

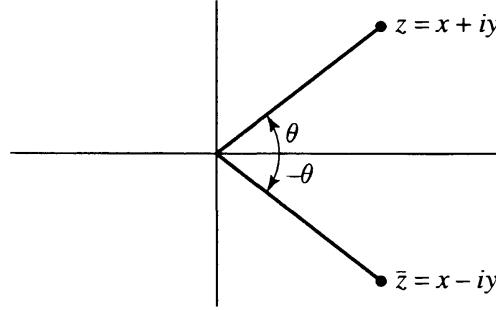
$$\begin{aligned} (z_1 z_2) z_3 &= z_1 (z_2 z_3), & (\text{associative law}) \\ z_1 z_2 &= z_2 z_1, & (\text{commutative law}) \\ z_1 (z_2 + z_3) &= z_1 z_2 + z_1 z_3. & (\text{distributive law}) \end{aligned}$$

With respect to algebraic operations, complex numbers behave the same as real numbers. Algebraic manipulations are performed on complex numbers using the usual laws of algebra, together with the special rule $i^2 = -1$.

Every complex number $z \neq 0$ has a multiplicative inverse $1/z$, which is given explicitly by

$$\frac{1}{z} = \frac{x - iy}{x^2 + y^2}, \quad z = x + iy \in \mathbb{C}, \quad z \neq 0.$$

Thus for instance, the multiplicative inverse of i is $1/i = -i$.



The **complex conjugate** of a complex number $z = x + iy$ is defined to be $\bar{z} = x - iy$. Geometrically, $\bar{z}$ is the reflection of z in the x -axis. If we reflect twice, we return to z ,

$$\bar{\bar{z}} = z, \quad z \in \mathbb{C}.$$

Some other useful properties of complex conjugation are

$$\begin{aligned} \overline{z + w} &= \bar{z} + \bar{w}, & z, w \in \mathbb{C}, \\ \overline{zw} &= \bar{z}\bar{w}, & z, w \in \mathbb{C}, \\ |z| &= |\bar{z}|, & z \in \mathbb{C}, \\ |z|^2 &= z\bar{z}, & z \in \mathbb{C}. \end{aligned}$$

Each of these identities can be verified easily using the definition of $\bar{z}$ and $|z|$. The last formula above allows us to express $1/z$ in terms of the complex conjugate $\bar{z}$:

$$1/z = \bar{z}/|z|^2, \quad z \in \mathbb{C}, z \neq 0.$$

The real and imaginary parts of z can be recovered from z and $\bar{z}$, by

$$\begin{aligned} \operatorname{Re} z &= (z + \bar{z})/2, & z \in \mathbb{C}, \\ \operatorname{Im} z &= (z - \bar{z})/2i, & z \in \mathbb{C}. \end{aligned}$$

From $|zw|^2 = (zw)(\overline{zw}) = (z\bar{z})(w\bar{w}) = |z|^2|w|^2$, we obtain also

$$|zw| = |z||w|, \quad z, w \in \mathbb{C}.$$

A **complex polynomial of degree $n \geq 0$** is a function of the form

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0, \quad z \in \mathbb{C},$$

where $a_0, \dots, a_n$ are complex numbers, and $a_n \neq 0$. A key property of the complex numbers, not enjoyed by the real numbers, is that any polynomial with complex coefficients can be factored as a product of linear factors.

Fundamental Theorem of Algebra. *Every complex polynomial $p(z)$ of degree $n \geq 1$ has a factorization*

$$p(z) = c(z - z_1)^{m_1} \cdots (z - z_k)^{m_k},$$

where the z_j 's are distinct and $m_j \geq 1$. This factorization is unique, up to a permutation of the factors.

We will not prove this theorem now, but we will give several proofs later. Some remarks are in order.

The uniqueness of the factorization is easy to establish. The points $z_1, \dots, z_k$ are uniquely characterized as the **roots** of $p(z)$, or the **zeros** of $p(z)$. These are the points where $p(z) = 0$. The integer m_j is characterized as the unique integer m with the property that $p(z)$ can be factored as $(z - z_j)^m q(z)$ where $q(z)$ is a polynomial satisfying $q(z_j) \neq 0$.

For the proof of the existence of the factorization, one proceeds by induction on the degree n of the polynomial. The crux of the matter is to find a point z_1 such that $p(z_1) = 0$. With a root z_1 in hand, one easily factors $p(z)$ as a product $(z - z_1)q(z)$, where $q(z)$ is a polynomial of degree $n - 1$. (See the exercises.) The induction hypothesis allows one to factor $q(z)$ as a product of linear factors, and this yields the factorization of $p(z)$. Thus the fundamental theorem of algebra is equivalent to the statement that every complex polynomial of degree $n \geq 1$ has a zero.

Example. The polynomial $p(x) = x^2 + 1$ with real coefficients cannot be factored as a product of linear polynomials with real coefficients, since it does not have any real roots. However, the complex polynomial $p(z) = z^2 + 1$ has the factorization

$$z^2 + 1 = (z - i)(z + i),$$

corresponding to the two complex roots $\pm i$ of $z^2 + 1$.

Exercises for I.1

1. Identify and sketch the set of points satisfying:

- | | |
|---------------------------------|--|
| (a) $ z - 1 - i = 1$ | (f) $0 < \operatorname{Im} z < \pi$ |
| (b) $1 < 2z - 6 < 2$ | (g) $-\pi < \operatorname{Re} z < \pi$ |
| (c) $ z - 1 ^2 + z + 1 ^2 < 8$ | (h) $ \operatorname{Re} z < z $ |
| (d) $ z - 1 + z + 1 \leq 2$ | (i) $\operatorname{Re}(iz + 2) > 0$ |
| (e) $ z - 1 < z $ | (j) $ z - i ^2 + z + i ^2 < 2$ |

2. Verify from the definitions each of the identities (a) $\overline{z + w} = \bar{z} + \bar{w}$, (b) $\overline{zw} = \bar{z}\bar{w}$, (c) $|\bar{z}| = |z|$, (d) $|z|^2 = z\bar{z}$. Draw sketches to illustrate (a) and (c).

3. Show that the equation $|z|^2 - 2 \operatorname{Re}(\bar{a}z) + |a|^2 = \rho^2$ represents a circle centered at a with radius ρ .
4. Show that $|z| \leq |\operatorname{Re} z| + |\operatorname{Im} z|$, and sketch the set of points for which equality holds.
5. Show that $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$. Show that

$$|z + w|^2 = |z|^2 + |w|^2 + 2 \operatorname{Re}(z\bar{w}).$$

Use this to prove the triangle inequality $|z + w| \leq |z| + |w|$.

6. For fixed $a \in \mathbb{C}$, show that $|z - a|/|1 - \bar{a}z| = 1$ if $|z| = 1$ and $1 - \bar{a}z \neq 0$.
7. Fix $\rho > 0$, $\rho \neq 1$, and fix $z_0, z_1 \in \mathbb{C}$. Show that the set of z satisfying $|z - z_0| = \rho|z - z_1|$ is a circle. Sketch it for $\rho = \frac{1}{2}$ and $\rho = 2$, with $z_0 = 0$ and $z_1 = 1$. What happens when $\rho = 1$?
8. Let $p(z)$ be a polynomial of degree $n \geq 1$ and let $z_0 \in \mathbb{C}$. Show that there is a polynomial $h(z)$ of degree $n - 1$ such that $p(z) = (z - z_0)h(z) + p(z_0)$. In particular, if $p(z_0) = 0$, then $p(z) = (z - z_0)h(z)$.
9. Find the polynomial $h(z)$ in the preceding exercise for the following choices of $p(z)$ and z_0 : (a) $p(z) = z^2$ and $z_0 = i$, (b) $p(z) = z^3 + z^2 + z$ and $z_0 = -1$, (c) $p(z) = 1 + z + z^2 + \cdots + z^m$ and $z_0 = -1$.
10. Let $q(z)$ be a polynomial of degree $m \geq 1$. Show that any polynomial $p(z)$ can be expressed in the form

$$p(z) = h(z)q(z) + r(z),$$

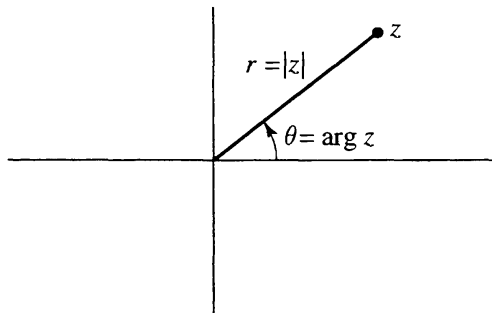
where $h(z)$ and $r(z)$ are polynomials and the degree of the remainder $r(z)$ is strictly less than m . *Hint.* Proceed by induction on the degree of $p(z)$. The resulting method is called the **division algorithm**.

11. Find the polynomials $h(z)$ and $r(z)$ in the preceding exercise for $p(z) = z^n$ and $q(z) = z^2 - 1$.

2. Polar Representation

Any point $(x, y) \neq (0, 0)$ in the plane can be described by polar coordinates r and θ , where $r = \sqrt{x^2 + y^2}$ and θ is the angle subtended by (x, y) and the x -axis. The angle θ is determined only up to adding an integral multiple of 2π . The Cartesian coordinates x, y are recovered from the polar coordinates r, θ by

$$\begin{cases} x = r \cos \theta, \\ y = r \sin \theta. \end{cases}$$



If we write the polar representation in complex notation, we obtain

$$(2.1) \quad z = x + iy = r(\cos \theta + i \sin \theta).$$

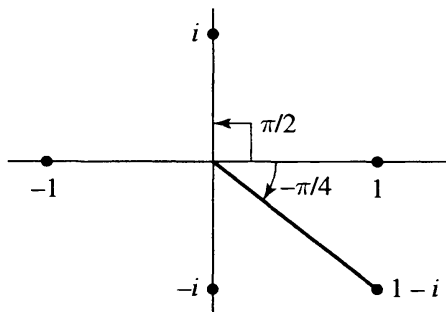
Here $r = |z|$ is the modulus of z . We define the **argument** of z to be the angle θ , and we write

$$\theta = \arg z.$$

Thus $\arg z$ is a multivalued function, defined for $z \neq 0$. The **principal value** of $\arg z$, denoted by $\text{Arg } z$, is specified rather arbitrarily to be the value of θ that satisfies $-\pi < \theta \leq \pi$. The values of $\arg z$ are obtained from $\text{Arg } z$ by adding integral multiples of 2π :

$$\arg z = \{\text{Arg } z + 2\pi k : k = 0, \pm 1, \pm 2, \dots\}, \quad z \neq 0.$$

Example. The principal value of $\arg i$ is $\text{Arg } i = \pi/2$. The principal value of $\arg(1 - i)$ is $\text{Arg}(1 - i) = -\pi/4$.



It will be convenient to introduce the notation

$$(2.2) \quad e^{i\theta} = \cos \theta + i \sin \theta.$$

From (2.1) we obtain

$$z = re^{i\theta}, \quad r = |z|, \quad \theta = \arg z.$$

This representation is called the **polar representation** of z . The sine and cosine functions are 2π -periodic, that is, they satisfy $\sin(\theta + 2\pi m) = \sin \theta$, $\cos(\theta + 2\pi m) = \cos \theta$. Thus the various choices of $\arg z$ yield the same value for $e^{i\theta}$,

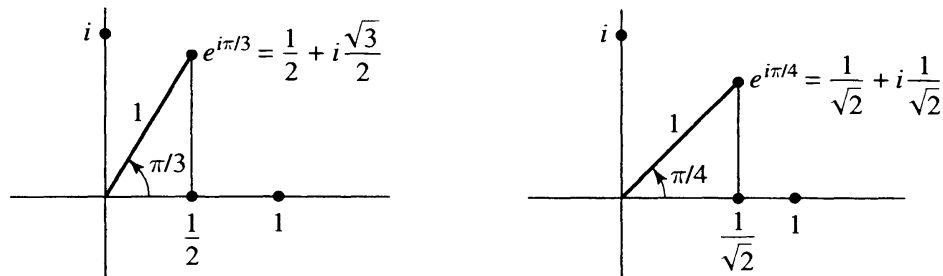
$$e^{i(\theta + 2\pi m)} = e^{i\theta}, \quad m = 0, \pm 1, \pm 2, \dots$$

Example. Some common complex exponentials are

$$e^{i\pi} = -1, \quad e^{i\pi/2} = i, \quad e^{i\pi/3} = \frac{1 + \sqrt{3}i}{2}, \quad e^{i\pi/4} = \frac{1 + i}{\sqrt{2}}.$$

Also note that

$$e^{2\pi mi} = 1, \quad m = 0, \pm 1, \pm 2, \dots$$



Several useful identities satisfied by the exponential function are

$$(2.3) \quad |e^{i\theta}| = 1,$$

$$(2.4) \quad \overline{e^{i\theta}} = e^{-i\theta},$$

$$(2.5) \quad 1/e^{i\theta} = e^{-i\theta}.$$

The identity (2.3) is equivalent to the trigonometric identity $\cos^2 \theta + \sin^2 \theta = 1$, while (2.4) follows from $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$.

One of the most important properties of the exponential function is the **addition formula**

$$(2.6) \quad e^{i(\theta+\varphi)} = e^{i\theta} e^{i\varphi}, \quad -\infty < \theta, \varphi < \infty.$$

In view of the definition (2.2), this is equivalent to

$$\cos(\theta + \varphi) + i \sin(\theta + \varphi) = (\cos \theta + i \sin \theta)(\cos \varphi + i \sin \varphi).$$

Multiplying out the right-hand side and equating real and imaginary parts, we obtain the equivalent pair of identities

$$(2.7) \quad \begin{cases} \cos(\theta + \varphi) = \cos \theta \cos \varphi - \sin \theta \sin \varphi, \\ \sin(\theta + \varphi) = \cos \theta \sin \varphi + \sin \theta \cos \varphi, \end{cases}$$

which are the addition formulae for sine and cosine. Thus the addition formula (2.6) for the complex exponential is a compact form of the addition formulae (2.7) for the sine and cosine functions, and it is much easier to remember!

The properties (2.4), (2.5), (2.6) of the exponential function correspond respectively to the following properties of the argument function:

$$(2.8) \quad \arg \bar{z} = -\arg z,$$

$$(2.9) \quad \arg(1/z) = -\arg z,$$

$$(2.10) \quad \arg(z_1 z_2) = \arg z_1 + \arg z_2,$$

where each formula is understood to hold modulo adding integral multiples of 2π . To establish (2.8) and (2.9), note that if the polar representation of z is $re^{i\theta}$, then the polar representation of $\bar{z}$ is $re^{-i\theta}$, and that of $1/z$ is $(1/r)e^{-i\theta}$. For (2.10), write $z_1 = r_1e^{i\theta_1}$, $z_2 = r_2e^{i\theta_2}$, and use the addition formula to obtain the polar form of z_1z_2 ,

$$z_1z_2 = r_1r_2e^{i\theta_1}e^{i\theta_2} = r_1r_2e^{i(\theta_1+\theta_2)}.$$

The addition formula (2.6) can be used to derive formulae for $\cos(n\theta)$ and $\sin(n\theta)$ in terms of $\cos\theta$ and $\sin\theta$. Write

$$\cos(n\theta) + i\sin(n\theta) = e^{in\theta} = (e^{i\theta})^n = (\cos\theta + i\sin\theta)^n,$$

expand the right-hand side, and equate real and imaginary parts. This yields expressions for $\cos(n\theta)$ and $\sin(n\theta)$ that are polynomials in $\cos\theta$ and $\sin\theta$. These identities are known as **de Moivre's formulae**. For instance, by equating $\cos(3\theta) + i\sin(3\theta)$ to

$$(\cos\theta + i\sin\theta)^3 = \cos^3\theta - 3\cos\theta\sin\theta + i(3\cos^2\theta\sin\theta - \sin^3\theta)$$

and taking real and imaginary parts, we obtain

$$\begin{aligned}\cos(3\theta) &= \operatorname{Re}(\cos\theta + i\sin\theta)^3 = \cos^3\theta - 3\cos\theta\sin^2\theta, \\ \sin(3\theta) &= \operatorname{Im}(\cos\theta + i\sin\theta)^3 = 3\cos^2\theta\sin\theta - \sin^3\theta.\end{aligned}$$

A complex number z is an **n th root** of w if $z^n = w$. Thus the n th roots of w are precisely the zeros of the polynomial $z^n - w$ of degree n . Since this polynomial has degree n , w has at most n n th roots. If $w \neq 0$, then w has exactly n n th roots, and these are determined as follows. First express w in polar form,

$$w = \rho e^{i\varphi}.$$

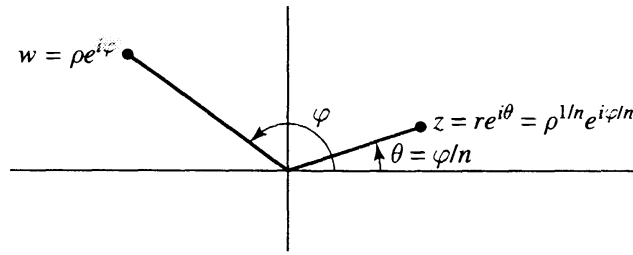
The equation $z^n = w$ becomes

$$r^n e^{in\theta} = \rho e^{i\varphi}.$$

Thus $r^n = \rho$ and $n\theta = \varphi + 2\pi k$ for some integer k . This leads to the explicit solutions

$$\begin{aligned}r &= \rho^{1/n}, \\ \theta &= \frac{\varphi}{n} + \frac{2\pi k}{n}, \quad k = 0, 1, 2, \dots, n-1,\end{aligned}$$

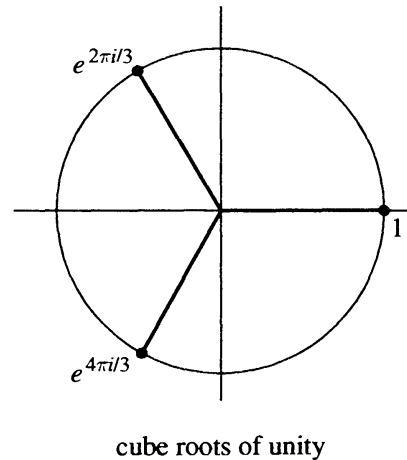
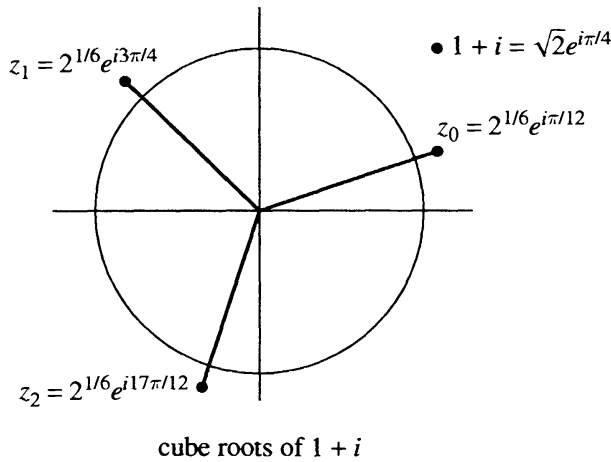
where we take the usual positive root of ρ . Since these n roots are distinct, and there are at most n n th roots, this list includes all the n th roots of w . Other values of k do not give different roots, since any other integer k leads to a value of θ that is obtained from the above list by adding an integral multiple of 2π . Graphically, the roots are distributed in equal arcs on the circle centered at 0 of radius $|w|^{1/n}$.



Example. To find and plot the square roots of $4i$, first express $4i$ in polar form $\rho e^{i\varphi}$. Here $\rho = |4i| = 4$ and $\varphi = \arg(4i) = \pi/2$. One root is given by $\sqrt{\rho} e^{i\varphi/2} = 2e^{i\pi/4}$. The other is $2e^{i(\pi/4+\pi)} = -2e^{i\pi/4}$. In Cartesian form, the roots are $\sqrt{4i} = \pm(\sqrt{2} + \sqrt{2}i)$.

Example. To find and plot the cube roots of $1 + i$, express $1 + i$ in polar form as $\sqrt{2} e^{i\pi/4}$. The polar form of the three cube roots is given by

$$2^{1/6} e^{i(\pi/12 + 2k\pi/3)}, \quad k = 0, 1, 2.$$



The n th roots of 1 are also called the **n th roots of unity**. They are given explicitly by

$$\omega_k = e^{2\pi i k/n}, \quad 0 \leq k \leq n-1.$$

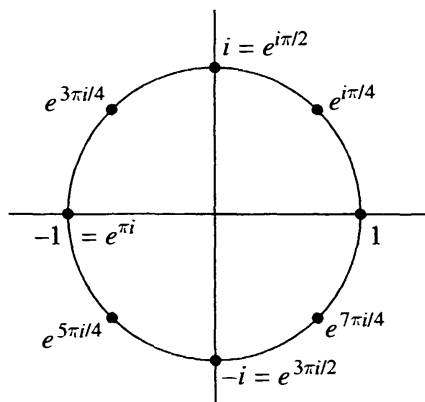
Graphically, they are situated at equal intervals around the unit circle in the complex plane. Thus the two square roots of unity are $e^0 = 1$ and $e^{i\pi} = -1$.

The procedure for finding the n th roots of $w \neq 0$ can be rephrased in terms of the n th roots of unity. We express $w = \rho e^{i\varphi/n}$ in polar form as above. One root is given by $z_0 = \rho^{1/n} e^{i\varphi/n}$. The others are found by multiplying z_0 by the n th roots of unity:

$$z_k = z_0 \omega_k = \rho^{1/n} e^{i\varphi/n} e^{2\pi i k/n}, \quad 0 \leq k \leq n-1.$$

Exercises for I.2

- Express all values of the following expressions in both polar and cartesian coordinates, and plot them.



The eight eighth roots of unity

- (a) $\sqrt{i}$ (c) $\sqrt[4]{-1}$ (e) $(-8)^{1/3}$ (g) $(1+i)^8$
 (b) $\sqrt{i-1}$ (d) $\sqrt[4]{i}$ (f) $(3-4i)^{1/8}$ (h) $\left(\frac{1+i}{\sqrt{2}}\right)^{25}$
2. Sketch the following sets:
 (a) $|\arg z| < \pi/4$ (c) $|z| = \arg z$
 (b) $0 < \arg(z-1-i) < \pi/3$ (d) $\log |z| = -2 \arg z$
3. For a fixed complex number b , sketch the curve $\{e^{i\theta} + be^{-i\theta} : 0 \leq \theta \leq 2\pi\}$. Differentiate between the cases $|b| < 1$, $|b| = 1$ and $|b| > 1$. *Hint.* First consider the case $b > 0$, and then reduce the general case to this case by a rotation.
4. For which n is i an n th root of unity?
5. For $n \geq 1$, show that
 (a) $1 + z + z^2 + \cdots + z^n = (1 - z^{n+1})/(1 - z)$, $z \neq 1$,
 (b) $1 + \cos \theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin(n + \frac{1}{2})\theta}{2 \sin \theta/2}$.
6. Fix $n \geq 1$. Show that the n th roots of unity $\omega_0, \dots, \omega_{n-1}$ satisfy:
 (a) $(z - \omega_0)(z - \omega_1) \cdots (z - \omega_{n-1}) = z^n - 1$,
 (b) $\omega_0 + \cdots + \omega_{n-1} = 0$,
 (c) $\omega_0 \cdots \omega_{n-1} = (-1)^{n-1}$,
 (d) $\sum_{j=0}^{n-1} \omega_j^k = \begin{cases} 0, & 1 \leq k \leq n-1, \\ n, & k = n. \end{cases}$
7. Fix $R > 1$ and $n \geq 1$, $m \geq 0$. Show that

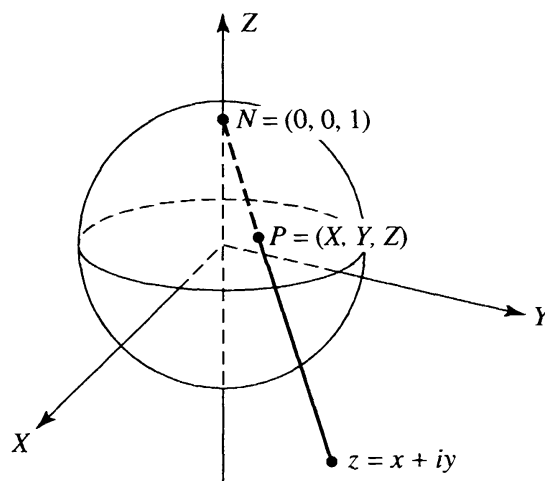
$$\left| \frac{z^m}{z^n + 1} \right| \leq \frac{R^m}{R^n - 1}, \quad |z| = R.$$

Sketch the set where equality holds. *Hint.* See (1.1).

8. Show that $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ and $\sin 2\theta = 2 \cos \theta \sin \theta$ using de Moivre's formulae. Find formulae for $\cos 4\theta$ and $\sin 4\theta$ in terms of $\cos \theta$ and $\sin \theta$.

3. Stereographic Projection

The **extended complex plane** is the complex plane together with the point at infinity. We denote the extended complex plane by $\mathbb{C}^*$, so that $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$. One way to visualize the extended complex plane is through stereographic projection. This is a function, or map, from the unit sphere in three-dimensional Euclidean space $\mathbb{R}^3$ to the extended complex plane, which is defined as follows. If $P = (X, Y, Z)$ is any point of the unit sphere other than the north pole $N = (0, 0, 1)$, we draw a straight line through N and P , and we define the **stereographic projection** of P to be the point $z = x + iy \sim (x, y, 0)$ where the straight line meets the coordinate plane $Z = 0$. The stereographic projection of the north pole N is defined to be ∞ , the point at infinity.



An explicit formula for the stereographic projection is derived as follows. We represent the line through P and N parametrically by $N + t(P - N)$, $-\infty < t < \infty$. The line meets the (x, y) -plane at a point $(x, y, 0)$ that satisfies

$$\begin{aligned} (x, y, 0) &= (0, 0, 1) + t[(X, Y, Z) - (0, 0, 1)] \\ &= (tX, tY, 1 + t(Z - 1)) \end{aligned}$$

for some parameter value t . Equating the third components, we obtain $0 = 1 + t(Z - 1)$, which allows us to solve for the parameter value t ,

$$t = 1/(1 - Z).$$

Equating the first two components and substituting this parameter value, we obtain equations for x and y in terms of X , Y , and Z ,

$$\begin{cases} x = tX = X/(1-Z), \\ y = tY = Y/(1-Z). \end{cases}$$

To solve for X, Y, Z in terms of x and y , we use the defining equation $X^2 + Y^2 + Z^2 = 1$ of the sphere. Multiplying this equation by t^2 and substituting $tX = x$, $tY = y$, $tZ = t-1$, we obtain $x^2 + y^2 + t^2 - 2t + 1 = t^2$, which becomes

$$t = \frac{1}{2}(|z|^2 + 1).$$

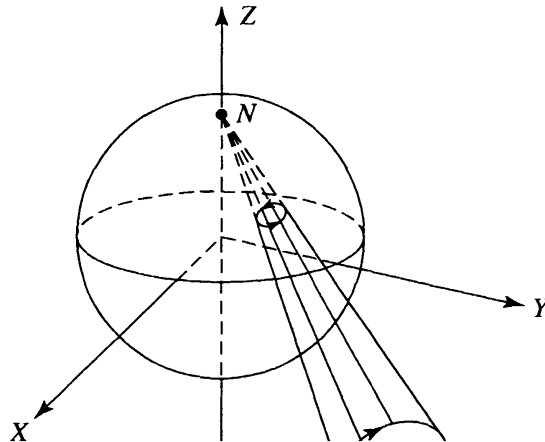
This yields

$$\begin{cases} X = 2x/(|z|^2 + 1), \\ Y = 2y/(|z|^2 + 1), \\ Z = 1 - 1/t = (|z|^2 - 1)/(|z|^2 + 1). \end{cases}$$

The point (X, Y, Z) of the sphere is determined uniquely by the point $z = x + iy$ of the plane. Thus the stereographic projection provides a one-to-one correspondence between points P of the sphere, except the north pole N , and points $z = x + iy$ of the complex plane.

Lines of longitude on the sphere correspond to straight lines in the plane through 0, while lines of latitude on the sphere correspond to circles centered at 0. As the radii of the circles tend to ∞ , the lines of latitude on the sphere tend to the north pole, so we are justified in making the north pole N correspond to the point at ∞ .

Theorem. *Under the stereographic projection, circles on the sphere correspond to circles and straight lines in the plane.*



To see this, we will use the fact that the locus of points in the plane satisfying a quadratic equation of the form

$$(3.1) \quad x^2 + y^2 + ax + by + c = 0$$

is either a circle, a point, or empty. This can be seen by completing the square and rewriting (3.1) as $(x + a/2)^2 + (y + b/2)^2 = (a^2 + b^2)/4 - c$. The three cases correspond respectively to whether $(a^2 + b^2)/4 - c$ is strictly positive, zero, or strictly negative.

We begin with a circle on the sphere, and we express it as the intersection of the sphere and a plane $AX + BY + CZ = D$. The stereographic projection of the circle then consists of points $z = x + iy$ that satisfy

$$A \frac{2x}{|z|^2 + 1} + B \frac{2y}{|z|^2 + 1} + C \frac{|z|^2 - 1}{|z|^2 + 1} = D.$$

We rewrite this as

$$(3.2) \quad (C - D)(x^2 + y^2) + 2Ax + 2By - (C + D) = 0.$$

If $C = D$, the locus of (3.2) is a straight line. If $C \neq D$, then we divide by $C - D$, and the equation (3.2) has the form (3.1). Being the projection of a circle on the sphere, the locus cannot be a point or empty, so it must be a circle in the plane.

The argument is reversible. Every circle in the plane is the locus of solutions of an equation of the form

$$x^2 + y^2 + A'x + B'y + D' = 0.$$

Define A, B, C, D so that $2A = A'$, $2B = B'$, $C - D = 1$, $-(C + D) = D'$, and the corresponding set on the sphere is the intersection of the sphere with the plane $AX + BY + CZ = D$. The intersection cannot be empty or a point; hence it is a circle on the sphere. Similarly, every straight line in the plane is the locus of solutions of an equation of the form $A'x + B'y = D'$, which also determines a plane via $2A = A'$, $2B = B'$, $C = D = D'/2$, and this plane meets the sphere in a circle through the north pole.

Since straight lines in the plane correspond to circles on the sphere through the north pole, it is convenient to regard a straight line in the complex plane as a **circle through ∞** . With this convention the theorem asserts simply that stereographic projection maps circles on the sphere to circles in the extended complex plane.

Exercises for I.3

1. Sketch the image under the spherical projection of the following sets on the sphere: (a) the lower hemisphere $Z \leq 0$, (b) the polar cap $\frac{3}{4} \leq Z \leq 1$, (c) lines of latitude $X = \sqrt{1 - Z^2} \cos \theta$, $Y = \sqrt{1 - Z^2} \sin \theta$, for Z fixed and $0 \leq \theta \leq 2\pi$, (d) lines of longitude $X = \sqrt{1 - Z^2} \cos \theta$, $Y = \sqrt{1 - Z^2} \sin \theta$, for θ fixed and $-1 \leq Z \leq 1$.

- (e) the spherical cap $A \leq X \leq 1$, with center lying on the equator, for fixed A . Separate into cases, according to various ranges of A .
2. If the point P on the sphere corresponds to z under the stereographic projection, show that the antipodal point $-P$ on the sphere corresponds to $-1/\bar{z}$.
 3. Show that as z traverses a small circle in the complex plane in the positive (counterclockwise) direction, the corresponding point P on the sphere traverses a small circle in the negative (clockwise) direction with respect to someone standing at the center of the circle and with body outside the sphere. (Thus the stereographic projection is orientation reversing, as a map from the sphere with orientation determined by the unit outer normal vector to the complex plane with the usual orientation.)
 4. Show that a rotation of the sphere of 180° about the X -axis corresponds under stereographic projection to the inversion $z \mapsto 1/\bar{z}$ of $\mathbb{C}$.
 5. Suppose $(x, y, 0)$ is the spherical projection of (X, Y, Z) . Show that the product of the distances from the north pole N to (X, Y, Z) and from N to $(x, y, 0)$ is 2. What is the situation when (X, Y, Z) lies on the equator of the sphere?
 6. We define the **chordal distance** $d(z, w)$ between two points $z, w \in \mathbb{C}^*$ to be the length of the straight line segment joining the points P and Q on the unit sphere whose stereographic projections are z and w , respectively. (a) Show that the chordal distance is a metric, that is, it is symmetric, $d(z, w) = d(w, z)$; it satisfies the triangle inequality $d(z, w) \leq d(z, \zeta) + d(\zeta, w)$; and $d(z, w) = 0$ if and only if $z = w$. (b) Show that the chordal distance from z to w is given by

$$d(z, w) = \frac{2|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}}, \quad z, w \in \mathbb{C}.$$

(c) What is $d(z, \infty)$? *Remark.* The expression for $d(z, w)$ shows that infinitesimal arc length corresponding to the chordal metric is given by

$$d\sigma(z) = \frac{2 ds}{1 + |z|^2},$$

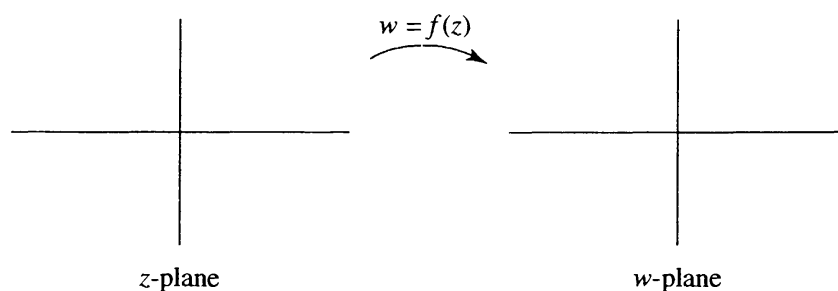
where $ds = |dz|$ is the usual Euclidean infinitesimal arc length. The infinitesimal arc length $d\sigma(z)$ determines another metric, the **spherical metric** $\sigma(z, w)$, on the extended complex plane. See Section IX.3.

7. Consider the sphere of radius $\frac{1}{2}$ in (X, Y, Z) -space, resting on the $(X, Y, 0)$ -plane, with south pole at the origin $(0, 0, 0)$ and north pole at $(0, 0, 1)$. We define a stereographic projection of the sphere onto the complex plane as before, so that corresponding points (X, Y, Z) and $z \sim (x, y, 0)$ lie on the same line through the north pole. Find the equations for $z = x + iy$ in terms of X, Y, Z , and the equations for X, Y, Z in terms of z . What is the corresponding formula for the chordal distance? *Note.* In this case, the equation of the sphere is $X^2 + Y^2 + (Z - \frac{1}{2})^2 = \frac{1}{4}$.

4. The Square and Square Root Functions

Real-valued functions of a real variable can be visualized by graphing them in the plane $\mathbb{R}^2$. The graph of a complex-valued function $f(z)$ of a complex variable z requires four (real) dimensions. Thus some techniques other than graphing in $\mathbb{R}^4$ must be developed for visualizing and understanding functions of a complex variable. One technique is to graph the modulus of the function $|f(z)|$ as a surface in three-dimensional space $\mathbb{R}^3$. Another is to graph separately the real and imaginary parts of $f(z)$ in $\mathbb{R}^3$.

We describe a different technique for gaining insight into the behavior of the function $f(z)$. We create two planes, a z -plane for the domain space and a w -plane for the range space. We then view $f(z)$ as a mapping from the z -plane to the w -plane, and we analyze how various geometric configurations in the z -plane are mapped by $w = f(z)$ to the w -plane. Which geometric configurations in the z -plane to consider depends very much on the specific function $f(z)$. To illustrate how this method works, we consider the simplest nontrivial function, the square function $w = z^2$.

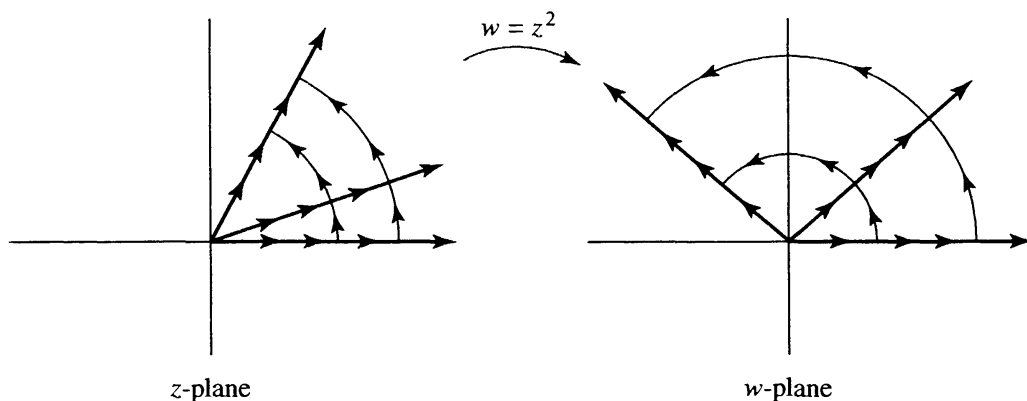


From the polar decomposition $w = z^2 = r^2 e^{2i\theta}$, we have

$$(4.1) \quad |w| = |z|^2,$$

$$(4.2) \quad \arg w = 2 \arg z.$$

Equation (4.1) shows that the circle $|z| = r_0$ in the z -plane is mapped to the circle $|w| = r_0^2$ in the w -plane. As $z = r_0 e^{i\theta}$ moves around the circle in the positive direction at constant angular velocity, the image $w = r_0^2 e^{2i\theta}$ moves around the image circle, in the same direction but at double the



angular velocity. As z makes one complete loop, the image w makes two complete loops around the image circle.

Equation (4.2) shows that a ray $\{\arg z = \theta_0\}$ issuing from the origin in the z -plane is mapped to a ray in the w -plane of twice the angle. As z traverses the ray from the origin to ∞ at constant speed, the value w traverses the image ray from 0 to ∞ , starting slowly and increasing its speed. The positive real axis in the z -plane, which is a ray with angle 0, is mapped to the positive real axis in the w -plane by the usual rule $x \mapsto x^2$. As z traverses the ray $\{\arg z = \pi/4\}$, the image w traverses the positive imaginary axis, and as z traverses the positive imaginary axis, the image w traverses the negative axis. As the rays in the z -plane sweep out the first quadrant, the image rays in the w -plane sweep out the upper half-plane, and as the rays in the z -plane sweep out the second quadrant, the image rays in the w -plane sweep out the lower half-plane. Eventually, we reach the ray along the negative real axis in the z -plane, which is mapped again to the positive real axis in the w -plane, and as we continue, the behavior is repeated in the lower half of the z -plane.

Now we turn to the problem of finding an inverse function for $w = z^2$. Every point $w \neq 0$ is hit by exactly two values of z , the two square roots $\pm\sqrt{w}$. In order to define an inverse function, we must restrict the domain in the z -plane so that values w are hit by only one z . There are many ways of doing this, and we proceed somewhat arbitrarily as follows.

Note that as rays sweep out the open right half of the z -plane, with the angle of the ray increasing from $-\pi/2$ to $\pi/2$, the image rays under $w = z^2$ sweep out the entire w -plane except for the negative axis, with the angle of the ray increasing from $-\pi$ to π . This leads us to draw a **slit**, or **branch cut**, in the w -plane along the negative axis from $-\infty$ to 0, and to define the inverse function on the **slit plane** $\mathbb{C} \setminus (-\infty, 0]$. Every value w in the slit plane is the image of exactly two z -values, one in the (open) right half-plane $\{\operatorname{Re} z > 0\}$, the other in the left half-plane $\{\operatorname{Re} z < 0\}$. Thus there are two possibilities for defining a (continuous) inverse function on the slit plane. We refer to each determination of the inverse function as a **branch** of the inverse. One branch $f_1(w)$ of the inverse function is defined by declaring that $f_1(w)$ is the value z such that $\operatorname{Re} z > 0$ and $z^2 = w$. Then $f_1(w)$ maps

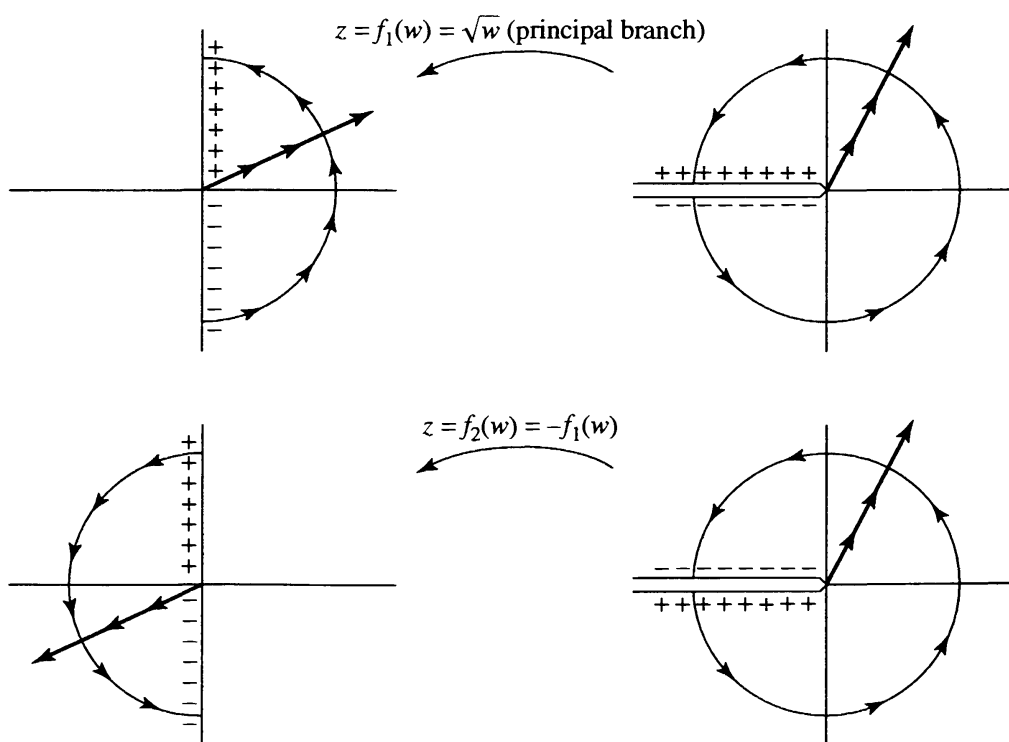
the slit plane $\mathbb{C} \setminus (-\infty, 0]$ onto the right half-plane $\{\operatorname{Re} z > 0\}$, and it forms an inverse for z^2 on that half-plane. To specify $f_1(w)$ explicitly, express $w = \rho e^{i\varphi}$ where φ lies in the range $-\pi < \varphi < \pi$, and then

$$f_1(w) = \sqrt{\rho} e^{i\varphi/2}, \quad w = \rho e^{i\varphi}, \quad -\pi < \varphi < \pi.$$

The function $f_1(w)$ is called the **principal branch of $\sqrt{w}$** . It is expressed in terms of the principal branch of the argument function as

$$f_1(w) = |w|^{1/2} e^{i(\operatorname{Arg} w)/2}, \quad w \in \mathbb{C} \setminus (-\infty, 0].$$

As w approaches a point $-r$ on the negative real axis $(-\infty, 0)$ from above, the values $f_1(w)$ approach the value $i\sqrt{r}$ on the positive imaginary axis. We express this by writing $f_1(-r + i0) = i\sqrt{r}$. Similarly, as w approaches $-r$ from below, the values $f_1(w)$ approach the value $-i\sqrt{r}$ on the negative imaginary axis, that is, $f_1(-r - i0) = -i\sqrt{r}$. The branch cut $(-\infty, 0]$ in the w -plane can be regarded as having two edges, and the function $f_1(w)$ extends continuously to each edge. The top edge, labeled “+” in the figure, is mapped to the positive imaginary axis by $f_1(w)$, and the bottom edge, labeled “−”, is mapped to the negative imaginary axis by $f_1(w)$.

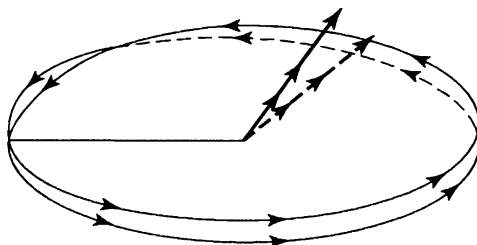


We use the other value of $\sqrt{w}$ to define a second branch $f_2(w)$ of the inverse function $\sqrt{w}$. For this we use a second copy of the w -plane, as in the figure. On this sheet the second branch of $\sqrt{w}$ is defined by $f_2(w) = -f_1(w)$. This branch maps the slit plane onto the left half-plane $\{\operatorname{Re} z < 0\}$. As w approaches a point $-r$ on the negative axis $(-\infty, 0)$ from above, the values $f_2(w)$ approach the value $-i\sqrt{r}$ on the negative imaginary axis,

and as w approaches $-r$ from below, the values $f_2(w)$ approach the value $+i\sqrt{r}$ on the positive imaginary axis. Again we think of the slit as having two edges, though on this sheet the top edge is mapped to the negative imaginary axis and the bottom edge is mapped to the positive imaginary axis. Further, we have

$$f_1(-r + i0) = i\sqrt{r} = f_2(-r - i0), \quad f_1(-r - i0) = -i\sqrt{r} = f_2(-r + i0).$$

This leads us to the idea of constructing a surface to represent the inverse function by gluing together the edges where the functions $f_1(w)$ and $f_2(w)$ coincide. We glue the top edge of the branch cut on the sheet corresponding to $f_1(w)$ to the bottom edge of the branch cut on the sheet corresponding to $f_2(w)$, and similarly for the remaining two edges, to obtain a two-sheeted surface. Since the values of $f_1(w)$ and $f_2(w)$ coincide on the edges we have glued together, they determine a function $f(w)$ defined on the two-sheeted surface, with values in the z -plane that move continuously with w .

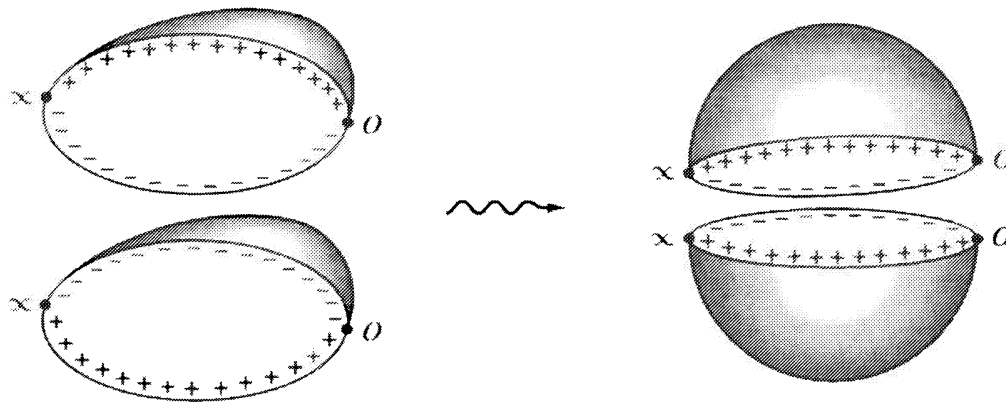


Since each sheet of the surface is a copy of the slit w -plane, we may think of the sheets as “lying over” the w -plane. Each $w \in \mathbb{C} \setminus \{0\}$ corresponds to exactly two points on the surface. The function $f(w)$ on the surface represents the multivalued function $\sqrt{w}$ in the sense that the values of $\sqrt{w}$ are precisely the values assumed by $f(w)$ at the points of the surface lying over w .

The surface we have constructed is called the **Riemann surface of $\sqrt{w}$** . The surface is essentially a sphere with two punctures corresponding to 0 and ∞ . One way to see this is to note that the function $f(w)$ maps the surface one-to-one onto the z -plane punctured at 0. Another way to see this is to deform the surface by prying open each sheet at the slit, opening it to a hemisphere, and then joining the two hemispheres along the slit edges to form a sphere with two punctures corresponding to the endpoints 0 and ∞ of the slits.

Exercises for I.4

- Sketch each curve in the z -plane, and sketch its image under $w = z^2$.
 - $|z - 1| = 1$
 - $x = 1$
 - $y = 1$
 - $y = x + 1$
 - $y^2 = x^2 - 1, x > 0$
 - $y = 1/x, x \neq 0$
- Sketch the image of each curve in the preceding problem under the principal branch of $w = \sqrt{z}$, and also sketch, on the same grid but



in a different color, the image of each curve under the other branch of $\sqrt{z}$.

3. (a) Give a brief description of the function $z \mapsto w = z^3$, considered as a mapping from the z -plane to the w -plane. (Describe what happens to w as z traverses a ray emanating from the origin, and as z traverses a circle centered at the origin.) (b) Make branch cuts and define explicitly three branches of the inverse mapping. (c) Describe the construction of the Riemann surface of $z^{1/3}$.
4. Describe how to construct the Riemann surfaces for the following functions: (a) $w = z^{1/4}$, (b) $w = \sqrt{z-i}$, (c) $w = (z-1)^{2/5}$. *Remark.* To describe the Riemann surface of a multivalued function, begin with one sheet for each branch of the function, make branch cuts so that the branches are defined continuously on each sheet, and identify each edge of a cut on one sheet to another appropriate edge so that the function values match up continuously.

5. The Exponential Function

We extend the definition of the exponential function to all complex numbers z by defining

$$e^z = e^x \cos y + ie^x \sin y, \quad z = x + iy \in \mathbb{C}.$$

Since $e^{iy} = \cos y + i \sin y$, this is equivalent to

$$e^z = e^x e^{iy}, \quad z = x + iy.$$

This identity is simply the polar representation of e^z ,

$$(5.1) \quad |e^z| = e^x,$$

$$(5.2) \quad \arg e^z = y.$$

If z is real ($y = 0$), the definition of e^z agrees with the usual exponential function e^x . If z is imaginary ($x = 0$), the definition agrees with the definition of $e^{i\theta}$ given in Section 2.

A fundamental property of the exponential function is that it is periodic. The complex number λ is a **period** of the function $f(z)$ if $f(z + \lambda) = f(z)$ for all z for which $f(z)$ and $f(z + \lambda)$ are defined. The function $f(z)$ is **periodic** if it has a nonzero period. Since $\sin x$ and $\cos y$ are periodic functions with period 2π , the function e^z is periodic with period $2\pi i$:

$$e^{z+2\pi i} = e^z, \quad z \in \mathbb{C}.$$

In fact, $2\pi ik$ is a period of e^z for any integer k .

Another fundamental property of the exponential function is the **addition formula**

$$(5.3) \quad e^{z+w} = e^z e^w, \quad z, w \in \mathbb{C}.$$

To check this, let $z = x + iy$ and $w = u + iv$. Then

$$e^{z+w} = e^{x+u} e^{i(y+v)} = e^x e^u e^{iy} e^{iv} = e^z e^w,$$

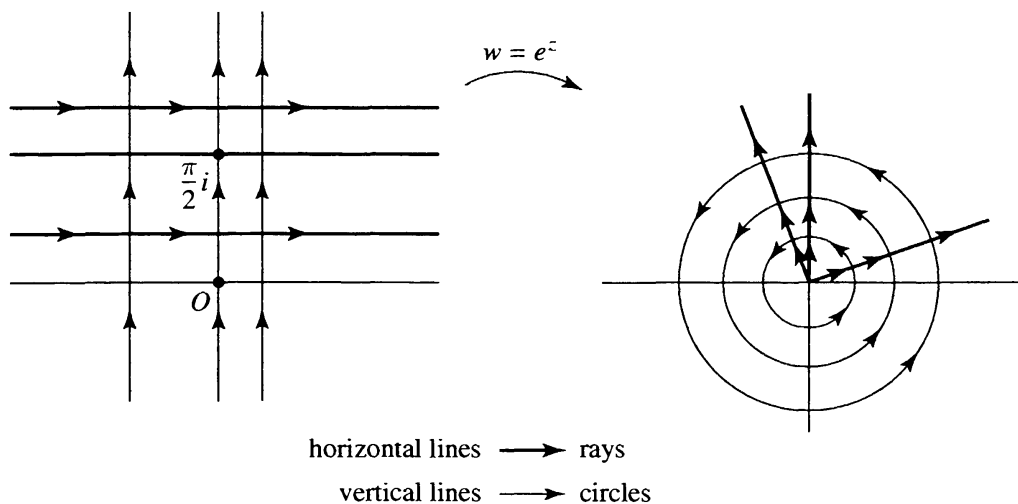
where we have used the addition formulae for e^x and $e^{i\theta}$.

From the addition formula (5.3) we have $e^z e^{-z} = e^0 = 1$. Consequently, the inverse of e^z is e^{-z} ,

$$1/e^z = e^{-z}, \quad z \in \mathbb{C}.$$

To understand the exponential function better, we view $w = e^z$ as a mapping from the z -plane to the w -plane. If we restrict the exponential function to the real line $\mathbb{R}$, we obtain the usual exponential function $x \mapsto e^x$, $-\infty < x < \infty$, which maps the real line $\mathbb{R}$ to the positive real axis $(0, \infty)$. The equation (5.2) shows that an arbitrary horizontal line $x + iy_0$, $-\infty < x < \infty$, is mapped to the curve $e^x e^{iy_0}$, $-\infty < x < \infty$, which is a ray issuing from the origin at angle y_0 . If we move the horizontal line up, the angle subtended by the ray increases, and the image ray is rotated in the positive (counterclockwise) direction. As we move the horizontal line upwards from the x -axis at $y_0 = 0$ to height $y_0 = \pi/2$, the image rays sweep out the first quadrant in the w -plane. The horizontal line at height $y_0 = \pi/2$ is mapped to the positive imaginary axis, the horizontal line of height $y_0 = \pi$ is mapped to the negative real axis, and when we reach the horizontal line of height $y_0 = 2\pi$, the image rays have swept out the full w -plane and returned to the positive real axis. The picture then repeats itself periodically. Each point in the w -plane, except $w = 0$, is hit infinitely often, by a sequence of z -values spaced at equal intervals of length 2π along a vertical line.

While the images of horizontal lines are rays issuing from the origin, the images of vertical lines are circles centered at the origin. The equation (5.1) shows that the image of the vertical line $x_0 + iy$, $-\infty < y < \infty$, is a circle in the w -plane of radius e^{x_0} . As z traverses the vertical line, the value w wraps infinitely often around the circle, completing one turn each time $y = \operatorname{Im} z$ increases by 2π .



Exercises for I.5

1. Calculate and plot e^z for the following points z :

(a) 0	(c) $\pi(i-1)/3$	(e) $\pi i/m$, $m = 1, 2, 3, \dots$
(b) $\pi i + 1$	(d) $37\pi i$	(f) $m(i-1)$, $m = 1, 2, 3, \dots$
2. Sketch each of the following figures and its image under the exponential map $w = e^z$. Indicate the images of horizontal and vertical lines in your sketch.
 - (a) the vertical strip $0 < \operatorname{Re} z < 1$,
 - (b) the horizontal strip $5\pi/3 < \operatorname{Im} z < 8\pi/3$,
 - (c) the rectangle $0 < x < 1$, $0 < y < \pi/4$,
 - (d) the disk $|z| \leq \pi/2$,
 - (e) the disk $|z| \leq \pi$,
 - (f) the disk $|z| \leq 3\pi/2$.
3. Show that $e^{\bar{z}} = \overline{e^z}$.
4. Show that the only periods of e^z are the integral multiples of $2\pi i$, that is, if $e^{z+\lambda} = e^z$ for all z , then λ is an integer times $2\pi i$.

6. The Logarithm Function

For $z \neq 0$ we define $\log z$ to be the multivalued function

$$\begin{aligned} \log z &= \log |z| + i \arg z \\ &= \log |z| + i \operatorname{Arg} z + 2\pi i m, \quad m = 0, \pm 1, \pm 2, \dots \end{aligned}$$

The values of $\log z$ are precisely the complex numbers w such that $e^w = z$. To see this, we plug in and compute. If $w = \log |z| + i \operatorname{Arg} z + 2\pi i m$, then

$$e^w = e^{\log |z|} e^{i \operatorname{Arg} z} e^{2\pi i m} = |z| e^{i \operatorname{Arg} z} = z,$$

where we have used the identities $e^{\log r} = r$ for $r > 0$ and $e^{2\pi im} = 1$. On the other hand, suppose that $w = u + iv$ is an arbitrary complex number such that $e^w = z$. Then the polar representation of z is $z = re^{iv}$, where $r = |z| = e^u$. Thus $u = \log |z|$, and v is a value of $\arg z$, so that $v = \text{Arg } z + 2\pi m$ for some integer m .

Recall that the principal value $\text{Arg } z$ of $\arg z$ is the value θ satisfying $-\pi < \theta \leq \pi$. We define the **principal value of $\log z$** to be

$$(6.1) \quad \text{Log } z = \log |z| + i \text{Arg } z, \quad z \neq 0.$$

Thus $\text{Log } z$ is a single-valued inverse for e^w , with values in the horizontal strip $-\pi < \text{Im } w \leq \pi$. Once we know the principal value of $\log z$, we obtain all values by

$$\log z = \text{Log } z + 2\pi im, \quad m = 0, \pm 1, \pm 2, \dots$$

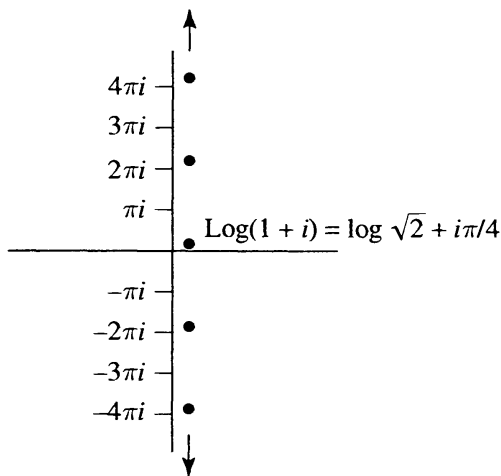
Example. The values of $\log(1 + i)$ are given by

$$\begin{aligned} \log(1 + i) &= \log |1 + i| + i \arg(1 + i) \\ &= \log \sqrt{2} + i\pi/4 + 2\pi im, \quad m = 0, \pm 1, \pm 2, \dots \end{aligned}$$

The principal value is

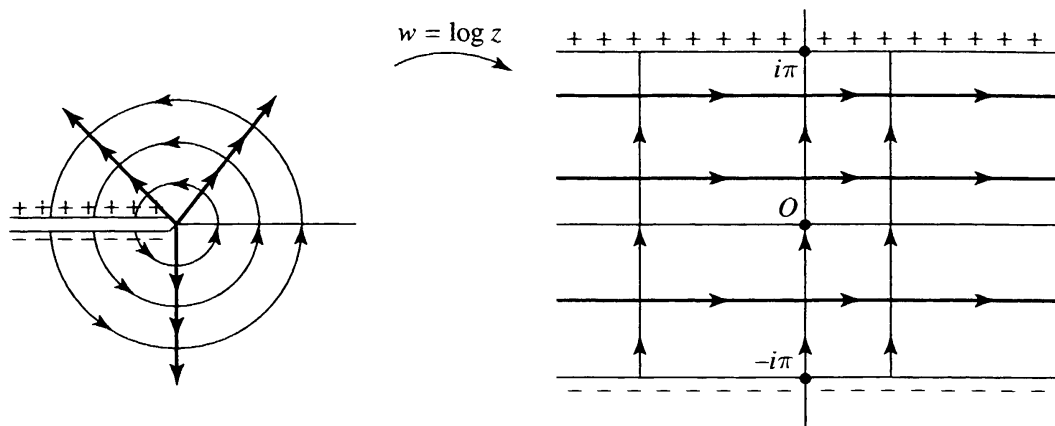
$$\text{Log}(1 + i) = \log \sqrt{2} + i\pi/4.$$

The values form a vertical two-tailed sequence of equally spaced points.



Now we regard $w = \text{Log } z$ as a map from the slit z -plane $\mathbb{C} \setminus (-\infty, 0]$ to the w -plane. Since the exponential function maps horizontal lines to rays issuing from the origin, its inverse, the logarithm function, maps rays issuing from the origin to horizontal lines. In fact, formula (6.1) shows that the ray $\{\text{Arg } z = \theta_0\}$ is mapped onto the horizontal line $\{\text{Im } w = \theta_0\}$. As z traverses the ray from 0 to ∞ , the image w traverses the entire horizontal line from left to right. As θ_0 increases between $-\pi$ and π , the rays sweep out the slit plane $\mathbb{C} \setminus (-\infty, 0]$, and the image lines fill out a horizontal strip

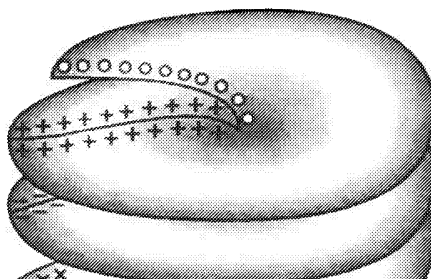
$\{-\pi < \operatorname{Im} w < \pi\}$ in the w -plane. Similarly, the formula (6.1) shows that the image of a punctured circle $\{|z| = r, -\pi < \arg z < \pi\}$ is the vertical interval $\{\operatorname{Re} w = \log |z|, -\pi < \operatorname{Im} z < \pi\}$, where the vertical line $\{\operatorname{Re} w = \log |z|\}$ meets the horizontal strip.



As with the inverse $\sqrt{z}$ of z^2 , we can represent the multivalued function $\log z$ as a single-valued function on a Riemann surface spread over the z -plane, with one sheet for each branch of the function. The construction is as follows. This time we have infinitely many branches $f_m(z)$ of the logarithm function, defined for $z \neq 0$ and given explicitly by

$$f_m(z) = \operatorname{Log} z + 2\pi im, \quad -\infty < m < \infty.$$

For each branch, we take a copy of the complex plane and slit it along the negative real axis as before, to obtain a copy S_m of the slit plane $\mathbb{C} \setminus (-\infty, 0]$. We regard the function $f_m(z)$ as defined on the m th sheet S_m . Since the values of $f_m(z)$ at the top edge of the slit on S_m match the values of $f_{m+1}(z)$ at the bottom edge of the slit on S_{m+1} , we glue together these two edges. We do this for each m , and we obtain a surface resembling a spiral stairway leading infinitely far both up and down. The composite function $f(z)$ defined to be $f_m(z)$ on the m th sheet is then continuously defined on the surface. It represents the total function $\log z$, in the sense that the values of $\log z$ are precisely the values of $f(z)$ at the points of the surface that correspond to z .



Exercises for I.6

- Find and plot $\log z$ for the following complex numbers z . Specify the principal value. (a) 2, (b) i , (c) $1 + i$, (d) $(1 + i\sqrt{3})/2$.
- Sketch the image under the map $w = \text{Log } z$ of each of the following figures.
 - the right half-plane $\text{Re } z > 0$,
 - the half-disk $|z| < 1$, $\text{Re } z > 0$,
 - the unit circle $|z| = 1$,
 - the slit annulus $\sqrt{e} < |z| < e^2$, $z \notin (-e^2, -\sqrt{e})$,
 - the horizontal line $y = e$,
 - the vertical line $x = e$.
- Define explicitly a continuous branch of $\log z$ in the complex plane slit along the negative imaginary axis, $\mathbb{C} \setminus [0, -i\infty)$.
- How would you make a branch cut to define a single-valued branch of the function $\log(z + i - 1)$? How about $\log(z - z_0)$?

7. Power Functions and Phase Factors

Let α be an arbitrary complex number. We define the power function z^α to be the multivalued function

$$z^\alpha = e^{\alpha \log z}, \quad z \neq 0.$$

Thus the values of z^α are given by

$$\begin{aligned} z^\alpha &= e^{\alpha[\log |z| + i \text{Arg } z + 2\pi im]} \\ &= e^{\alpha \text{Log } z} e^{2\pi i \alpha m}, \quad m = 0, \pm 1, \pm 2, \dots \end{aligned}$$

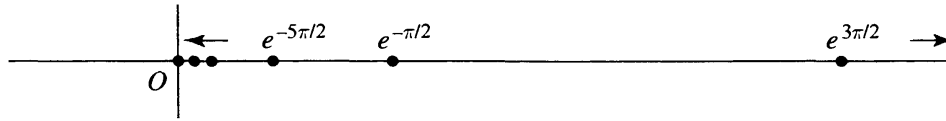
The various values of z^α are obtained from the principal value $e^{\alpha \text{Log } z}$ by multiplying by the integral powers $(e^{2\pi i \alpha})^m$ of $e^{2\pi i \alpha}$. If α is itself an integer, then $e^{2\pi i \alpha} = 1$, and the function z^α is single-valued, the usual power function. If $\alpha = 1/n$ for some integer n , then the integral powers $e^{2\pi i m/n}$ of $e^{2\pi i/n}$ are exactly the n th roots of unity, and the values of $z^{1/n}$ are the n n th roots of z discussed earlier (Section 2).

Example. The values of i^i are given by

$$e^{i \log i} = e^{-\text{Arg } i - 2\pi m} = e^{-\pi/2} e^{-2\pi m}, \quad m = 0, \pm 1, \pm 2, \dots$$

The values form a two-tailed sequence of positive real numbers, accumulating at 0 and at $+\infty$. Similarly, the values of i^{-i} are given by

$$e^{i \log(-i)} = e^{-\text{Arg}(-i) - 2\pi k} = e^{\pi/2} e^{-2\pi k}, \quad k = 0, \pm 1, \pm 2, \dots$$



Danger! If we multiply the values of i^i by those of i^{-i} , we obtain infinitely many values $e^{2\pi n}$, $-\infty < n < \infty$. Thus

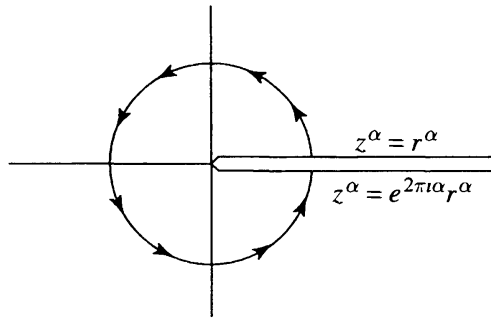
$$(i^i)(i^{-i}) \neq i^0 = 1,$$

and the usual algebraic rules do not apply to power functions when they are multivalued.

If α is not an integer, we cannot define z^α on the entire complex plane in such a way that the values move continuously with z . To define the function continuously, we must again make a branch cut. We could make the cut along the negative real axis, but this time let us make the cut along the positive real axis, from 0 to $+\infty$. We define a continuous branch of z^α on the slit plane $\mathbb{C} \setminus [0, \infty)$ explicitly by

$$w = r^\alpha e^{i\alpha\theta}, \quad \text{for } z = re^{i\theta}, \quad 0 < \theta < 2\pi.$$

At the top edge of the slit, corresponding to $\theta = 0$, we have the usual power function $r^\alpha = e^{\alpha \operatorname{Log} r}$. At the bottom edge of the slit, corresponding to $\theta = 2\pi$, we have the function $r^\alpha e^{2\pi i\alpha}$. If we fix r and let θ increase from 0 to 2π , $z = re^{i\theta}$ starts at the top edge of the slit and proceeds around a circle, ending at the bottom edge of the slit. As z describes this circle, the values $w = r^\alpha e^{i\theta\alpha}$ move continuously, starting from r^α at the top edge of the slit and ending at $r^\alpha e^{2\pi i\alpha}$ at the bottom edge. Thus the values of this branch of z^α on the bottom edge are $e^{2\pi i\alpha}$ times the values at the top edge. The multiplier $e^{2\pi i\alpha}$ is called the **phase factor** of z^α at $z = 0$.



If we continue any other choice $w = r^\alpha e^{i\alpha(\theta+2\pi m)}$ of z^α around the same circle, the values of w move continuously from $r^\alpha e^{2\pi i\alpha m}$ at the top edge of the slit to $r^\alpha e^{i\alpha(2\pi+2\pi m)} = r^\alpha e^{2\pi i\alpha m} e^{2\pi i\alpha}$ at the bottom edge. Again the final w -value is the phase factor $e^{2\pi i\alpha}$ times the initial w -value.

The same analysis shows that the function $(z - z_0)^\alpha$ has a phase factor of $e^{2\pi i\alpha}$ at $z = z_0$, in the sense that if any branch of $w = (z - z_0)^\alpha$ is continued around a full circle centered at z_0 in the positive direction, the final w -value is $e^{2\pi i\alpha}$ times the initial w -value. This can be seen by making

a change of variable $\zeta = z - z_0$. Further, this result does not change if we multiply $(z - z_0)^\alpha$ by any (single-valued) function. We state the result formally for emphasis.

Phase Change Lemma. *Let $g(z)$ be a (single-valued) function that is defined and continuous near z_0 . For any continuously varying branch of $(z - z_0)^\alpha$ the function $f(z) = (z - z_0)^\alpha g(z)$ is multiplied by the phase factor $e^{2\pi i \alpha}$ when z traverses a complete circle about z_0 in the positive direction.*

Example. If α is an integer, the phase factor of z^α at 0 is $e^{2\pi i \alpha} = 1$, in accord with the fact that z^α is single-valued.

Example. The phase factor of $\sqrt{z - z_0}$ at z_0 is $e^{\pi i} = -1$. As z traverses a circle about z_0 , the values of $f(z) = \sqrt{z - z_0}$ return to $-f(z)$. The phase factor of $1/\sqrt{z_0 - z} = i/\sqrt{z - z_0}$ at z_0 is also -1 .

Example. The function $\sqrt{z(1 - z)}$ has two branch points, at 0 and at 1. At $z = 0$, each branch of $\sqrt{1 - z}$ is single-valued, so the phase factor of each branch of $\sqrt{z(1 - z)}$ at $z = 0$ is the same as that of $\sqrt{z}$, which is -1 . Similarly, the phase factor of $\sqrt{z(1 - z)}$ at $z = 1$ is the same as that of $\sqrt{1 - z}$, which is -1 . Now suppose we draw a branch cut from 0 to 1 and consider the branch $f(z)$ of $\sqrt{z(1 - z)}$ that is positive on the top edge of the slit. As z traverses a small circle around 0, the values of $f(z)$ return to $-f(z)$ on the bottom edge of the slit, corresponding to the phase factor -1 at $z = 0$. As z traverses the bottom edge of the slit and returns to the top edge around a small circle at $z = 1$, the values of $-f(z)$ are again multiplied by the phase factor -1 . Thus the values of $f(z)$ return to the original positive value on the top edge of the slit when z traverses a dogbone path encircling both branch points. It follows that the branch $f(z)$ is a continuous single-valued function in the slit plane $\mathbb{C} \setminus [0, 1]$. Now we may proceed, in analogy with $\sqrt{z}$ and $\log z$, to define a Riemann surface for the function $\sqrt{z(1 - z)}$ that captures both branches of the function. We require two sheets, since there are two choices of branches for the function $\sqrt{z(1 - z)}$. On each sheet we make the same cut, to form two copies of $\mathbb{C} \setminus [0, 1]$. On one sheet we define $F(z)$ to be the branch $f(z)$ of $\sqrt{z(1 - z)}$ specified above, and on the other sheet we define $F(z)$ to be the other branch $-f(z)$ of $\sqrt{z(1 - z)}$. The sheets are then joined by identifying edges of the slits in such a way that $F(z)$ extends continuously to the surface. In this case, the top edge of the slit $[0, 1]$ on one sheet is identified to the bottom edge of the slit on the other sheet, and the remaining two edges are identified, to form the two-sheeted Riemann surface of $\sqrt{z(1 - z)}$.

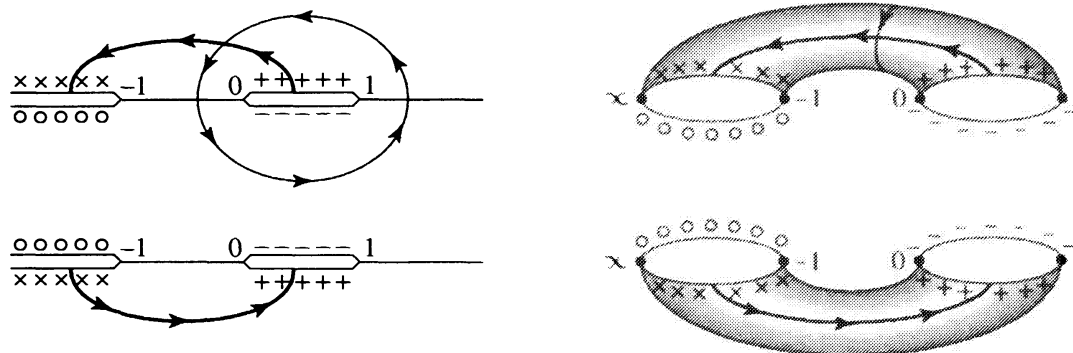
In constructing the Riemann surface of a multivalued function, the number of sheets always coincides with the number of branches of the function. However, the branch cuts can be made in many ways, as long as there are



surface with closed path

enough branch cuts so that each branch of the function can be defined continuously in the slit plane. For instance, the branch cuts for the function $f(x) = \sqrt{z(1-z)}$ could as well be made from $-\infty$ to 0 along the negative real axis and from $+1$ to $+\infty$ along the positive real axis. The branch cuts could also be made along more complicated paths from 0 to 1.

Example. Consider $\sqrt{z-1}/z$. We rewrite this as $\sqrt{z-1}\sqrt{z+1}/\sqrt{z}$. The function has three finite branch points, at 0 and ± 1 . We must also consider ∞ as a branch point, since there is a phase change corresponding to a phase factor -1 as z traverses a very large circle centered at 0. Each branch point has phase factor -1 , so any branch of the function returns to its original values when z traverse a path encircling two of the branch points. Thus it suffices to make two cuts, say $(-\infty, -1]$ and $[0, 1]$. Each branch of the function is continuous on $\mathbb{C} \setminus ((-\infty, -1] \cup [0, 1])$. Again top edges of slits on one sheet are identified to bottom edges of the others. The resulting surface is a torus (doughnut, or inner tube), with punctures corresponding to the branch points. What would happen if we were to make initially an additional branch cut along $[-1, 0]$, in addition to the two branch cuts above? The values of each branch at the top edge of the new cut would agree with the values of the same branch on the bottom edge. Consequently, we would identify the top and bottom edges of the slit $[-1, 0]$ on the same sheet, thereby effectively erasing the slits and arriving at the same doughnut surface.



Exercises for I.7

- Find all values and plot: (a) $(1+i)^i$, (b) $(-i)^{1+i}$, (c) $2^{-1/2}$, (d) $(1+i\sqrt{3})^{(1-i)}$.
- Compute and plot $\log[(1+i)^{2i}]$.

3. Sketch the image of the sector $\{0 < \arg z < \pi/6\}$ under the map $w = z^a$ for (a) $a = \frac{3}{2}$, (b) $a = i$, (c) $a = i + 2$. Use only the principal branch of z^a .
4. Show that $(zw)^a = z^a w^a$, where on the right we take all possible products.
5. Find i^{i^i} . Show that it does not coincide with $i^{i \cdot i} = i^{-1}$.
6. Determine the phase factors of the function $z^a(1-z)^b$ at the branch points $z = 0$ and $z = 1$. What conditions on a and b guarantee that $z^a(1-z)^b$ can be defined as a (continuous) single-valued function on $\mathbb{C} \setminus [0, 1]$?
7. Let $x_1 < x_2 < \cdots < x_n$ be n consecutive points on the real axis. Describe the Riemann surface of $\sqrt{(z-x_1) \cdots (z-x_n)}$. Show that for $n = 1$ and $n = 2$ the surface is topologically a sphere with certain punctures corresponding to the branch points and ∞ . What is it when $n = 3$ or $n = 4$? Can you say anything for general n ? (Any compact Riemann surface is topologically a sphere with handles. Thus a torus is topologically a sphere with one handle. For a given n , how many handles are there, and where do they come from?)
8. Show that $\sqrt{z^2 - 1/z}$ can be defined as a (single-valued) continuous function outside the unit disk, that is, for $|z| > 1$. Draw branch cuts so that the function can be defined continuously off the branch cuts. Describe the Riemann surface of the function.
9. Consider the branch of the function $\sqrt{z(z^3 - 1)(z + 1)^3}$ that is positive at $z = 2$. Draw branch cuts so that this branch of the function can be defined continuously off the branch cuts. Describe the Riemann surface of the function. To what value at $z = 2$ does this branch return if it is continued continuously once counterclockwise around the circle $\{|z| = 2\}$?
10. Consider the branch of the function $\sqrt{z(z^3 - 1)(z + 1)^3(z - 1)}$ that is positive at $z = 2$. Draw branch cuts so that this branch of the function can be defined continuously off the branch cuts. Describe the Riemann surface of the function. To what value at $z = 2$ does this branch return if it is continued continuously once counterclockwise around the circle $\{|z| = 2\}$?
11. Find the branch points of $\sqrt[3]{z^3 - 1}$ and describe the Riemann surface of the function.